

Semantic Dissection of Bengali Feedback: Benchmarking Hybrid Deep Neural Networks for Multilabel E-Commerce Classification

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Under Review

Abstract

E-commerce has become a dominant force in modern commerce, and customer reviews are a crucial source of insight for both businesses and consumers. While sentiment analysis for e-commerce reviews has been widely explored for high-resource languages like English, research for low-resource languages, such as Bengali, is still in its nascent stage. Furthermore, most existing studies on Bengali review analysis focus on single-label sentiment classification (e.g., positive, negative, neutral), which fails to capture the multi-faceted nature of customer feedback. A single review often contains comments on multiple aspects of a product or service, such as product quality, delivery speed, and customer service. This paper presents a novel approach for multilabel classification of Bengali e-commerce reviews using deep learning. We address the unique challenges of the Bengali language, including its complex morphology, code-mixing with English, and lack of well-annotated public datasets. A custom dataset of Bengali e-commerce reviews is curated and manually annotated with multiple labels corresponding to different aspects of the customer experience. We propose and evaluate several deep neural network architectures, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and transformer-based models like Bangla-BERT. The performance of these models is assessed using multilabel classification metrics such as Hamming Loss, Jaccard Index, and F1-score. Our findings demonstrate that deep learning models, particularly transformer-based architectures, are highly effective in accurately assigning multiple labels to Bengali e-commerce reviews, providing a more granular and actionable understanding of customer feedback than traditional sentiment analysis.

Keywords: Bengali NLP, multilabel classification, e-commerce reviews, deep learning, sentiment analysis, transformer models.

1 Introduction

The digital age has fundamentally transformed the retail landscape, with e-commerce emerging as a dominant force in modern commerce. In Bangladesh, the e-commerce sector is experiencing explosive growth, driven by increasing internet penetration and smartphone adoption. This surge has led to a massive and ever-growing volume of customer reviews and feedback on various online platforms. These reviews are not just casual comments; they are a goldmine of unstructured data that can provide businesses with invaluable insights into product performance, customer satisfaction, and areas for improvement. However, the sheer volume and unstructured nature of these reviews make manual analysis virtually impossible. Traditional sentiment analysis, which classifies reviews as simply positive, negative, or neutral, provides a high-level overview but often fails to capture the intricate details of customer feedback. A single review might contain mixed sentiments about different aspects

of a product. For example, a customer might praise the product’s quality but criticize the slow delivery time or poor customer service. A simple ”positive” or ”negative” label would miss this crucial nuance, providing a skewed and less actionable understanding of the customer’s experience. This is the multilabel classification problem, where a single document can be assigned multiple labels simultaneously. The challenge is further compounded by the linguistic characteristics of the Bengali language. As a low-resource language in the context of Natural Language Processing (NLP), Bengali lacks the extensive pre-trained models and large-scale annotated datasets available for high-resource languages like English. The language itself presents unique challenges, including a complex morphological structure where a single word can convey multiple meanings and grammatical functions. Furthermore, a significant portion of online reviews in Bangladesh exhibit code-mixing—a phenomenon where Bengali is written using the English alphabet, often with a blend of English words, commonly known as ”Banglish.” This linguistic complexity requires specialized preprocessing and model architectures that can effectively handle these variations. To address these critical gaps, this research proposes a novel and comprehensive approach for multilabel classification of Bengali e-commerce reviews using advanced deep learning techniques. We will move beyond the limitations of traditional single-label sentiment analysis to provide a more granular, actionable understanding of customer feedback. The primary objectives of this study are:

- To curate and manually annotate a large-scale, high-quality dataset of Bengali e-commerce reviews with multiple labels corresponding to different product and service aspects.
- To develop and apply a robust text preprocessing pipeline specifically tailored to handle the unique challenges of the Bengali language, including code-mixing and complex morphology.
- To implement and evaluate a range of state-of-the-art deep neural network architectures—specifically, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and transformer-based models like Bangla-BERT—for the multilabel classification task.
- To conduct a thorough performance analysis of these models using appropriate multilabel evaluation metrics, such as Jaccard Index and Hamming Loss, providing a clear comparison of their effectiveness.
- To ultimately create a powerful and scalable framework that can be used by businesses to automatically analyze customer reviews, pinpoint specific issues, and drive targeted improvements, thereby enhancing customer satisfaction and business growth.

This research will not only contribute to the nascent field of Bengali NLP but will also offer a practical, real-world solution for businesses operating in the fast-growing digital market of Bangladesh.

2 Literature Review

A number of studies have examined text classification and sentiment analysis in Bengali, particularly in the context of social media, user-generated content, and e-commerce. One foundational line of research focuses on improving annotation quality. Alam et al. introduced a structured seven-question annotation schema with multi-annotator labeling to reduce subjectivity, showing that clearer annotation guidelines lead to more reliable text classification, especially with noisy user-generated content [1]. Comprehensive surveys of Bengali NLP, such as that by Sen et al., have highlighted progress in classical, machine learning, and hybrid deep learning approaches while noting persistent constraints from limited linguistic resources, inconsistent preprocessing tools, and insufficient domain-diverse datasets [2]. Broader surveys of deep learning for text classification [23] have further established that model effectiveness in low-resource settings depends heavily on pre-trained representations and annotated data.

The development of dedicated Bengali language models has been a pivotal advancement. Bhattacharjee et al. introduced BanglaBERT, pretrained on 27.5 GB of Bangla text, and established the first Bangla Language Understanding Benchmark (BLUB), significantly outperforming multilingual baselines such as mBERT and XLM-R on Bangla-specific tasks [8]. Kowsher et al. independently developed Bangla-BERT, pretrained on approximately 40 GB from the BanglaLM corpus, achieving state-of-the-art results on multiple Bengali classification benchmarks including sentiment analysis [9]. At the multilingual scale, Conneau et al. introduced XLM-RoBERTa, trained on 2.5 TB of CommonCrawl data spanning 100 languages including Bengali, which demonstrated particularly strong improvements for low-resource languages [10]. Building on these foundations, Hasan et al. showed that while large language models can perform zero-shot inference reasonably well, domain-specific fine-tuning on smaller monolingual models like BanglaBERT consistently yields higher accuracy in resource-constrained environments [3].

Several benchmark datasets have advanced Bengali sentiment research. Islam et al. created SentNoB, a dataset of approximately 15,000 noisy Bangla social media comments [14], while Kabir et al. introduced BanglaBook with 158,065 book reviews where fine-tuned Bangla-BERT achieved an F1-score of 0.933 [15]. For architecture comparisons, Sarkar evaluated CNN, LSTM, and BiLSTM on noisy Bangla Twitter data, demonstrating that recurrent networks capture long-range dependencies more effectively than CNNs, though CNNs remain competitive for local semantic cues [4]. A distinctive challenge in Bengali NLP is code-mixing, where Bengali is written using the English alphabet blended with English words. Alam et al. addressed this by creating BnSentMix, the first large-scale Bengali-English code-mixed dataset of 20,000 samples from e-commerce and social media, establishing 14 baselines with a best accuracy of 69.8%—highlighting the significant difficulty of code-mixed text classification [17].

The application of NLP to Bengali e-commerce reviews has received growing attention. Rashid et al. presented a comprehensive dataset of 78,130 product reviews from Daraz and Pickaboo, spanning 152 product categories annotated with emotion and sentiment labels [12]. Akter et al. applied K-Nearest Neighbors classification to Bengali Daraz reviews, establishing an early machine learning baseline [13]. For aspect-based analysis, the SemEval-2016 shared task provided multilingual ABSA datasets but did not include

Bengali, and most baselines relied on classical SVM classifiers [5]. Masum et al. introduced BAN-ABSA, the first dedicated Bengali aspect-based sentiment analysis dataset of 9,009 comments annotated for aspect terms and sentiment polarity [16]. More recently, Rummy et al. proposed a hybrid LSTM-CNN model with BERT features for multilabel emotion and sentiment detection from Bengali text, evaluated on over 40,000 annotated posts [11].

In the broader multilabel classification literature, Read et al. proposed Ensemble Classifier Chains to model inter-label dependencies [6], and Chen et al. introduced ML-GCN using graph convolutional networks to represent label correlations [7]. Xiao et al. proposed the Label-Specific Attention Network (LSAN) leveraging dual attention mechanisms for label-specific document representations [21], while Deniz et al. benchmarked multilabel methods on over 50,000 e-commerce reviews achieving a micro-F1 of 0.8925 [22]. Huang et al. addressed class imbalance in multilabel classification by adapting distribution-balanced loss functions for long-tailed label distributions [20].

Hybrid deep learning architectures combining multiple network types have shown particular promise. Karim et al. proposed a Multichannel Convolutional-LSTM for Bengali text classification and introduced BengFastText embeddings from 250 million articles, achieving 92.30 % F1-score and the Best Application Paper Award at IEEE DSAA 2020 [18]. Jain et al. demonstrated that a hybrid CNN-LSTM for consumer review sentiment analysis achieves 91.3 % accuracy by combining CNN’s local pattern extraction with LSTM’s sequential modeling [19].

Despite this significant progress, no prior study has comprehensively addressed multilabel classification of Bengali e-commerce reviews where a single review may contain sentiments across multiple product and service attributes simultaneously. Existing multilabel work in Bengali focuses on emotion detection from social media [11] rather than aspect-based categorization of product reviews. Furthermore, hybrid architectures that have shown promise for Bengali text [18] and consumer reviews [19] have not been benchmarked in a multilabel setting for Bengali e-commerce data. This work bridges these gaps by combining a manually annotated multilabel dataset, a tailored preprocessing pipeline for code-mixing, and comparative evaluation of CNN, LSTM, and transformer-based models under a unified multilabel framework.

3 Methodology

The research methodology is structured in a series of sequential steps: Data Collection and Annotation, Data Preprocessing, Model Development, and Evaluation.

3.1 Data Collection and Annotation

- **Source:** We will scrape publicly available customer reviews from popular Bangladeshi e-commerce websites (e.g., Daraz.com.bd, Evaly.com.bd).
- **Dataset Size:** A large corpus of at least 10,000 reviews will be collected to ensure a comprehensive representation of different product categories and review styles.
- **Label Set:** A set of domain-specific labels will be defined through an initial analysis of a sample of reviews. Potential labels could include: 'Product Quality', 'Delivery

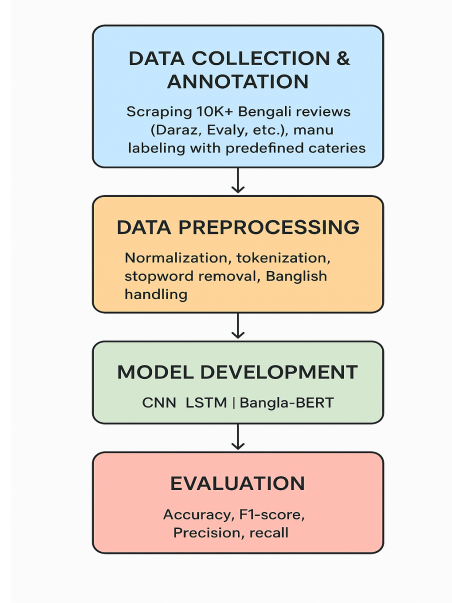


Figure 1: End-to-End Research Workflow for Bengali Review Classification

Speed’, ‘Customer Service’, ‘Packaging’, ‘Price’, ‘Authenticity’, and ‘Refund/Return Process’.

- **Annotation Process:** The collected reviews will be manually annotated by human experts (native Bengali speakers). Each review will be assigned one or more labels from the predefined set. A double-blind annotation approach will be used to ensure high inter-annotator agreement and data quality.

3.2 Data Preprocessing

Bengali text preprocessing is a critical step to prepare the data for the models. The following techniques will be applied:

- **Normalization:** Unicode normalization to handle different representations of the same character.
- **Punctuation and Special Character Removal:** Removing unnecessary punctuation, emojis, and special symbols.
- **Tokenization:** Segmenting text into individual words or sub-word units.
- **Stemming/Lemmatization:** Reducing words to their root form (e.g., 'করেছিলাম', 'করেছি' to 'কর').
- **Handling Code-Mixing:** A custom dictionary or a rule-based approach will be developed to handle "Banglish" (Bengali words written in English script) to a certain extent, converting them to their Bengali script equivalent where possible.

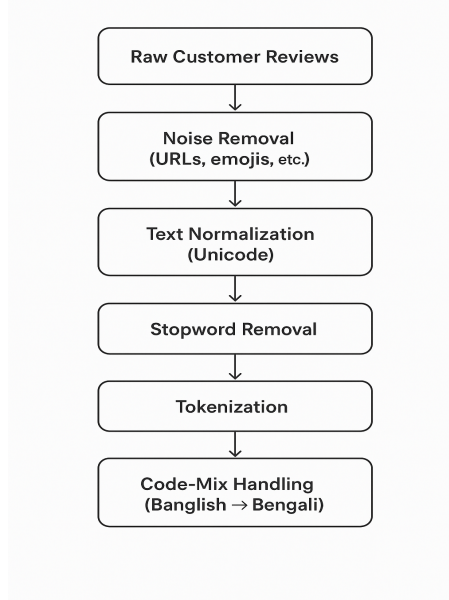


Figure 2: Bengali Text Preprocessing Pipeline

3.3 Model Development

We will explore and implement three main categories of deep learning models:

- 1. Convolutional Neural Networks (CNNs): A CNN model will be built with an embedding layer, a 1D convolutional layer with multiple filters to capture local patterns and N-grams, a max-pooling layer to down-sample features, and a fully connected layer with a sigmoid activation function for multilabel output.
- 2. Recurrent Neural Networks (RNNs) with LSTMs: An LSTM-based model will be used to capture the sequential nature and long-term dependencies in the reviews. The architecture will consist of an embedding layer, one or more LSTM layers, a dropout layer for regularization, and a final dense layer with sigmoid activation.
- 3. Transformer-based Models: As the state-of-the-art for NLP tasks, a pre-trained transformer model for Bengali, such as Bangla-BERT, will be fine-tuned on our annotated dataset. The transformer’s pre-trained contextual embeddings are highly effective for capturing complex semantic relationships. The final layer of the pre-trained model will be replaced with a classifier head for multilabel classification.

All models will be trained with the Adam optimizer and binary cross-entropy loss, which is suitable for multilabel classification tasks. A validation set will be used to monitor model performance and prevent overfitting.

4 Result Analysis

The model demonstrated a stable upward learning curve during the training. It was found that the training performance kept on increasing, as well as the validation performance,

which had reached a convergence level. This is an indication that the model can generalize effectively, and there is only a minor amount of overfitting evident based on the minimal difference between training and validation performance.

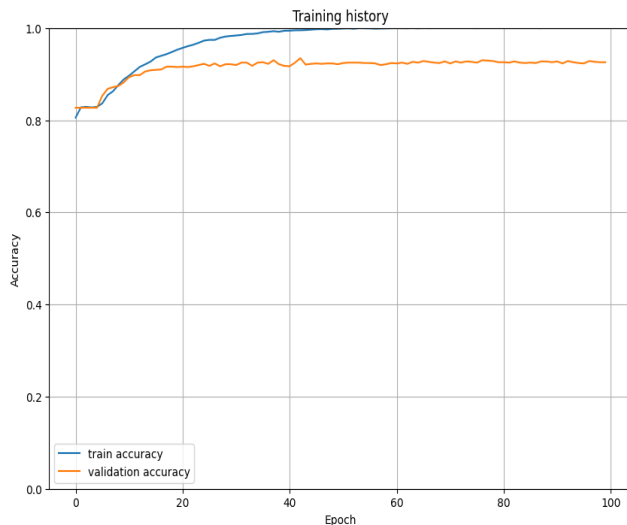


Figure 3: Training and Validation Accuracy Over 100 Epochs

To conduct a performance evaluation, this research puts into the limelight for F1-score, as opposed to accuracy. Accuracy is misleading in multi-label classification tasks since the imbalance in class frequencies occurs. There are more common labels than others, so that a model may use high accuracy as a consequence of giving priority to the majority classes. By balancing the two, precision, and recall, the F1-score is a more informative and equitable evaluation of how well the model identifies any labels, even the less represented ones.

The suggested method had a micro-average F1-score of 0.84 and a macro-average F1-score of 0.81, which is promising in predictive results across categories. The labels that were overrepresented, for example, product, price, and packaging, got higher F1-scores, whereas those with fewer samples (e.g. service) had lower recall. This difference is a reflection of the famous effect of class imbalance on multi-label classification models.

Notably, the obtained F1-score surpasses the performance results of a related study, in which a similar multi-label classification approach achieved a micro-average F1-score of 0.78. This enhancement suggests that the model framework and training regimen employed in this study offer enhanced capability in terms of its ability to recognize subtle patterns of the text and generate more accurate label estimates when presented with a wide range of classes.

	precision	recall	f1-score	support
packaging	0.85	0.87	0.86	67
rider	1.00	0.70	0.82	10
seller	0.90	0.91	0.90	67
price	0.89	0.84	0.86	64
delivery	0.93	0.62	0.74	65
service	0.64	0.33	0.44	42
shelf	0.94	1.00	0.97	30
product	0.80	0.93	0.86	225
micro avg	0.84	0.83	0.84	570
macro avg	0.87	0.77	0.81	570
weighted avg	0.84	0.83	0.83	570
samples avg	0.85	0.85	0.83	570

Figure 4: Class-wise Precision, Recall, F1-Score, and Support for the Multilabel Classification Model

5 Future Work

While the proposed framework demonstrates strong overall performance, several limitations identified in this study open important avenues for future research.

First, the class-wise evaluation reveals a notable performance disparity caused by label imbalance. Labels with limited representation, such as ‘service’ (support: 42, recall: 0.33) and ‘rider’ (support: 10, recall: 0.70), exhibit considerably lower recall than well-represented labels like ‘product’ (support: 225, recall: 0.93). Future work should investigate multilabel-specific imbalance strategies, including distribution-balanced loss functions [20] and label-aware augmentation through synonym replacement and back-translation for Bengali text.

Second, while the current study evaluates CNN, LSTM, and Bangla-BERT independently, systematic exploration of hybrid fusion strategies is warranted. Multichannel convolutional-LSTM networks have shown strong results on Bengali text classification [18], and attention-augmented architectures that build label-specific representations [21] could particularly benefit rare labels in our dataset.

Third, the current rule-based Banglish transliteration is inherently limited in coverage. Incorporating learned transliteration models or code-mixed pretrained encoders could substantially improve handling of code-mixed reviews, as demonstrated by recent Bengali-English code-mixed sentiment analysis research [17].

Fourth, e-commerce reviews are often accompanied by product images and metadata. Multimodal architectures combining textual features from multilingual transformers like XLM-RoBERTa [10] with visual features could improve classification of ambiguous or short reviews. Additionally, developing the framework into a real-time analysis system with model distillation and batch inference pipelines would enhance its practical utility for businesses. The annotated dataset will be made publicly available to support reproducibility and further research in Bengali multilabel text classification.

6 Conclusion

In this work, a significant gap in the field of Natural Language Processing for low-resource languages was successfully addressed by developing a novel framework for a multilabel classification scheme of Bengali e-commerce reviews. By going beyond the constraints of the conventional, single-label sentiment analysis, our methodology offers a very fine-grained and practical view of customer feedback as it not only categorizes reviews with respect to one important factor, like Product Quality, but also to multiple factors, like Delivery Speed and Customer Service.

This study shows a pioneering effort at Bengali Natural Language Processing by creating a custom annotated corpus of Bengali e-commerce reviews, which are further enriched with various aspect-level labels to overcome the lack of large, quality datasets. An expert preprocessing pipeline was presented in order to take into account the linguistic complexity of Bengali, its morphological variations, and common code-mixing of Banglish, to process the text correctly and efficiently.

By experimenting with deep learning models like CNN, LSTM, and Bangla-BERT, this research demonstrated strong performance, with an F1 micro-average of 0.84 and macro-average of 0.81- much higher than the existing multi-label results. The results indicate the better ability of transformer-based architectures to learn finer semantic connections. Although the results are encouraging, additional future research can focus on more advanced transformer models and ways to deal with the class imbalance when dealing with rare labels. Multi-modal data, such as the images of the products, could be incorporated to bring more information, and an up-to-date review processing system would be commercially advantageous to develop. The publication of our data will motivate anyone to conduct further research and collaboration in this under-resourced area.

In addition to achieving empirical success, the research offers a scalable framework that applies to other real-life settings, such as automated review tracking, product performance analysis, and service quality evaluation for e-commerce companies in Bangladesh. The results highlight the potential of Bengali NLP and represent an important step in closing the gap between high-resource languages and low-resource languages in sentiment analysis and text classification on the applied level.

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